Validity Challenges in Inferring AI Tool Behavior from Observational Pull Request Datasets

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***Abstract*—Observational datasets of AI-assisted pull requests face methodological challenges that can invalidate behavioral inferences. We identify four critical validity threats: agent attribution uncertainty, statistical clustering violations, measurement bias, and confounding effects. Using AIDev as a case study, we demonstrate how these threats manifest and establish methodological requirements for reliable inference from observational AI tool data.**

I. Introduction

Large-scale datasets of AI-assisted pull requests offer insights into real-world tool usage but face methodological challenges that can invalidate behavioral inferences. This study identifies and demonstrates validity threats in observational AI tool datasets, using AIDev (932,791 PRs) as a case study.

We identify four critical validity threats: (1) agent attribution reliability, (2) statistical clustering violations, (3) measurement bias, and (4) confounding effects. We demonstrate how these threats render behavioral inference unreliable.

Contribution: A methodological framework identifying validity requirements for reliable inference from observational AI tool datasets.

II. Validity Threats in Observational AI Tool Research

Observational AI tool datasets face four primary validity threats:

Attribution Reliability: Metadata and self-reports may misclassify tool usage due to incomplete records, mixed usage, or detection errors.

Statistical Clustering: Users, repositories, and organizations create correlated observations that violate independence assumptions, inflating significance tests.

Measurement Bias: Keyword-based outcome detection produces systematic errors, particularly across different languages and practices.

Confounding Effects: Developers select tools based on projects, experience, and context, making tool effects impossible to isolate without controls.

III. Empirical Demonstrations

A. Attribution Evidence

Table I shows extreme user count imbalances suggesting attribution problems.

TABLE I
Agent attribution patterns revealing reliability concerns

Agent	Users	PRs/User
OpenAI Codex	61,653	13.2
GitHub Copilot	379	133.1
Claude Code	1,643	3.1
Cursor	9,658	3.4

Such extreme disparities in contribution patterns suggest systematic labeling inconsistencies rather than genuine usage differences.

Critical Disclaimer: This paper does not present behavioral results. Any numerical examples illustrate how validity threats distort inference and should not be interpreted as evidence of agent behavior.

IV. Related Work

Recent code-generation studies emphasize functional correctness and security [2], while empirical work reports developer experiences with assistants such as Copilot [3]. Broader empirical software engineering has explored validity challenges in observational studies [5], [7]. Our methodological framework complements this work by identifying specific validity requirements for AI tool research.

V. Implications

For Researchers: Prioritize controlled experiments for behavioral inference. When using observational data, implement comprehensive validity checks and avoid causal claims.

For Practitioners: Interpret observational studies as suggestive rather than definitive. Focus on controlled evaluations for important decisions.

VI. Methodological Requirements

Valid behavioral inference requires:

Attribution Verification: Independent validation achieving ¿90% accuracy with balanced errors across groups.

Clustering Control: Multilevel modeling or cluster-robust methods accounting for user, repository, and organizational structure.

Measurement Validation: Manual validation ($n \geq 500$) across languages and contexts, with bias quantification.

Confounding Assessment: Collection of developer, project, and temporal covariates with appropriate analytical controls.

VII. Limitations

Validity Threats: Agent labels are unreliable, statistical tests are invalid due to clustering, and severe confounding prevents causal inference. This analysis cannot distinguish agent effects from artifacts.

Ethics. We analyze public data for methodological illustration only. This critique targets research methods, not individuals or organizations.

VIII. Conclusion

This study demonstrates validity threats that prevent reliable behavioral inference from observational AI tool datasets. We identify four critical threats and establish methodological requirements for valid inference. Without these safeguards, apparent patterns likely reflect methodological artifacts. This critique targets methodology, not researchers or datasets, aiming to improve research rigor in AI tool evaluation.

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References

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